

Satyam Kumar

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[LinkedIn](#) | [GitHub](#)

EDUCATION

Vivekananda Institute of Professional Studies – TC, GGSIPU

2023 – 2027

B.Tech (AI & ML) | CGPA: 7.6 / 10 (till 5th sem)

TECHNICAL SKILLS

Languages : Python, HTML, CSS

Data Analysis : NumPy, Pandas, Matplotlib, EDA & Data Preprocessing

ML / DL : Scikit-learn, TensorFlow (Basic), OpenCV

Tools : GitHub, Hugging Face Spaces, Google Colab, VS Code, Kaggle

Creative Tools: Photoshop, Premier Pro, Canva, Figma, Framer

Concepts : Machine Learning, Deep Learning, Computer Vision, Image Segmentation, NLP, OOP, UI/UX

PROJECTS

House Price Prediction System

[GitHub](#) | [Demo](#)

Python, Scikit-learn, Linear Regression, Gradio

- Developed an end-to-end house price prediction system using the Ames Housing dataset containing **2,900+ residential properties** and **80+ explanatory features**
- Achieved an average test-set prediction error of 5.8%, enabling reliable valuation across diverse housing profiles and market segments
- Featuring data preprocessing, feature engineering, model evaluation, and EDA, with an interactive user interface.
- Deployed an interactive Gradio application **with 20 configurable property attributes**, real-time price estimation, and **90% confidence** range generation

DeepFake Shield - Facial Manipulation Detection System

[GitHub](#) | [Demo](#)

TensorFlow, EfficientNetB4, FFT, GradCAM, Gradio, Hugging Face

- Developed a dual-branch deepfake detector using EfficientNetB4 and FFT spectrum analysis, achieving **77.6% classification accuracy** on 10K+ real and fake face images
- Implemented GradCAM explainability and deployed a real-time Gradio application on Hugging Face Spaces with prediction heatmaps and FFT visualization

Life Under Water - Marine Trash Detection System

[GitHub](#) | [Demo](#)

Python, YOLOv8, Gradio, Hugging Face

- Fine-tuned YOLOv8m-seg on TrashCan 1.0 dataset containing 7,212 underwater images across 16 classes; achieved **mAP50: 65.6% and 79% precision** over 75 epochs
- Identified severe class imbalance (59:1) and implemented class-specific confidence thresholds to improve recall on rare trash categories
- Deployed an end-to-end detection pipeline on Hugging Face with image/video inference and real-time detection dashboard aligned with UN SDG 14

CERTIFICATES

E & ICT Academy, IIT Kanpur — Digital Marketing - Comprehensive Studies

UiPath — Introduction to agentic automation

POSITIONS & ACTIVITIES

SM Head — Kalakriti

- Managed the society's social media presence and promoted events through engaging content.

Co-Head — No. 1 Delhi Air Sq'n (NCC)

- Managed cross-functional Social Media Team departments and streamlined operations

Member (GD) — 35MM

- Designed digital and print creatives to support branding and event promotions